From Environmental Chaoticity to Quantum Logical Error Rates: A Random-Matrix-Theory Pipeline for Surface-Code Noise Mapping

Problem Statement, Framework, Implementation, Preliminary Results, and a Roadmap Toward a PRX Quantum Submission

Working Technical Report

July 30, 2026

Abstract

We describe a computational pipeline that converts environmental sensor time-series data (light, distance, temperature, humidity) captured from an Arduino-based data-logger into Random Matrix Theory (RMT) spectral statistics, extracts a Brody chaoticity parameter $w \in [0, 1]$ characterizing the regularity/chaoticity of the local environment, and maps w onto a physical Pauli error probability and a resulting surface-code logical failure rate E_L. This report documents (i) the scientific problem statement and motivation, (ii) the full mathematical framework with every formula used, (iii) the software implementation, (iv) preliminary results obtained from real Arduino-captured sensor data, (v) a rigorous, self-critical evaluation of the current results against the standards expected by *Physical Review X Quantum* (PRX Quantum), and (vi) a phased roadmap of the technical and statistical work required before the study is ready for submission. The central finding to date is a strong, but currently statistically overconfident and physically uncalibrated, monotonic association between w and E_L; the roadmap in Section **??** lays out the concrete steps needed to convert this into a defensible, referee-proof scientific claim.

Contents

1 Problem Statement

1.1 Motivation

Quantum error correction (QEC) protocols, including the surface code, are designed and benchmarked under the standard assumption of a *stationary*, well-characterized physical noise channel acting on each qubit. In practice, however, the physical substrate of a quantum processor is embedded in a real laboratory environment whose ambient conditions — temperature, humidity, vibration, stray electromagnetic and optical fields — fluctuate on timescales ranging from milliseconds to hours. These fluctuations are a plausible, but rarely directly quantified, contributor to non-stationary and non-Markovian noise in near-term quantum hardware.

A natural diagnostic question is: *can the statistical "character" of ambient environmental fluctuations be quantified in a way that is predictive of the severity of the noise a nearby qubit would experience, and ultimately of the resulting logical error rate under a given QEC code?* Answering this requires (a) a principled way to characterize the regularity vs. chaoticity of an environmental time series, and (b) a mapping from that characterization to a physical noise model and a QEC performance metric.

1.2 Why Random Matrix Theory

Random Matrix Theory provides exactly this kind of principled, parameter-free characterization in the closely related context of quantum spectra: the nearest-neighbor spacing distribution of energy levels distinguishes *integrable* (regular) quantum systems, which exhibit Poissonian statistics, from *chaotic* systems, whose spectra exhibit Wigner–Dyson level repulsion consistent with the Gaussian Orthogonal Ensemble (GOE). The Brody distribution (Section **??**) interpolates continuously between these two limits via a single chaoticity parameter $w \in [0, 1]$.

This report investigates whether the same RMT machinery, applied not to a quantum Hamiltonian but to a *delay-embedded classical time series* (in the spirit of Takens-embedding / Singular Spectrum Analysis), yields a meaningful chaoticity indicator $w(t)$ for an ambient sensor environment, and whether $w(t)$ correlates with a downstream, physically motivated surface-code logical error rate $E_L(t)$.

1.3 Formal Problem Statement

Given: a multivariate environmental time series $\{x_c(t_k)\}$, $c \in \{\text{ldr}, \text{distance}, \text{temperature}, \text{humidity}\}$, sampled at discrete times t_k by an Arduino-based sensor logger.
Construct: a windowed estimator $\hat{w}(t)$ of the local RMT chaoticity of each channel, and a downstream estimator $\hat{E}_L(t)$ of the logical error rate of a distance-d surface code operating under a Pauli noise channel whose strength is set by $\hat{w}(t)$.
Determine: whether $\hat{w}(t)$ and $\hat{E}_L(t)$ are statistically and physically meaningfully associated, with rigorously quantified uncertainty, in a way that would constitute either (i) a genuine physical finding about environment-induced QEC degradation, or (ii) a validated general-purpose diagnostic *methodology* for such studies.

1.4 Current Status

As of this report, stages (i)–(iii) above have been implemented and executed on both synthetic and real Arduino-captured sensor data (Section **??**). The association between $\hat{w}$ and $\hat{E}_L$ is present and visually strong (Figure **??**), but — as detailed in Section **??** — several statistical and physical-modeling gaps currently prevent this from constituting a publishable claim at the standard required by PRX Quantum. Section **??** enumerates the concrete remaining work.

2 Framework Overview

2.1 Pipeline Architecture

The pipeline is organized into five sequential, independently testable stages, implemented as five Python classes (Section **??**):

1. **`DatasetLoader`** — loads and validates the Arduino CSV log (columns: `timestamp_ms`, `ldr`, `distance_cm`, `temperature_c`, `humidity_pct`); standardizes each channel to zero mean, unit variance.
2. **`HankelMatrixBuilder`** — slices each standardized channel into overlapping windows, embeds each window as a Hankel trajectory matrix, and forms a symmetric Wishart matrix whose real eigenvalue spectrum is extracted.
3. **`SpectralUnfolderAndBrodyFitter`** — unfolds the raw eigenvalue spectrum using a Legendre-polynomial expansion of the cumulative spectral density, extracts normalized nearest-neighbor spacings, and fits the Brody distribution to obtain $\hat{w}$.
4. **`SurfaceCodeNoiseMapper`** — maps $\hat{w}$ to a physical fault probability p_{phys}, constructs the corresponding single-qubit Pauli channel, and computes the logical error rate E_L of a distance-d surface code (via an analytical below-threshold scaling law, with an optional `stim` + `pymatching` circuit-level simulation fallback path).
5. **`PublicationVisualizer`** — produces APS/PRX-style figures: the Brody spacing-distribution fit, the dual-axis $w(t)/E_L(t)$ time series, and the w-vs-E_L correlation scatter.

Data flow:

$$\underbrace{x_c(t_k)}_{\text{raw sensor}} \xrightarrow{\text{standardize}} \hat{x}_c(t_k) \xrightarrow{\text{Hankel embed}} H \xrightarrow{\frac{1}{p}HH^{\top}} W \xrightarrow{\text{eigh}} \{\lambda_i\}$$

$$\{\lambda_i\} \xrightarrow{\text{Legendre unfold}} \{s_i\} \xrightarrow{\text{Brody fit}} \hat{w} \xrightarrow{p_0(1+\alpha\hat{w}^{\beta})} p_{\mathrm{phys}} \to \mathcal{E}(\rho) \to E_L$$

2.2 Design Rationale

Each modeling choice in the framework is intentionally modular so that any single stage can later be replaced with a more rigorously validated or empirically calibrated component without altering the rest of the pipeline. In particular:

- The RMT unfolding and Brody-fitting stages (2–3) are grounded in a long-established formalism from quantum chaos and nuclear physics (Brody *et al.*, 1981; Bohigas–Giannoni–Schmit conjecture; Mehta, *Random Matrices*), applied here to a delay-embedded classical signal in the spirit of Singular Spectrum Analysis / Takens embedding.
- The noise-mapping stage (4) currently uses *heuristic, illustrative* functional forms ($p_{\mathrm{phys}}(w)$ and the below-threshold $E_L(p_{\mathrm{phys}})$ scaling law) that are structurally standard in the QEC literature but whose *coefficients* (p_0, α, β, C) are not yet calibrated against real hardware or a first-principles noise model. This is

the single largest gap identified in Section **??**.

3 Mathematical Formulation

This section presents every formula used in the pipeline, in the order in which it is applied, together with a detailed explanation of its role and derivation context.

3.1 Stage 1: Standardization

For each sensor channel x with n samples,

$$\hat{x}_i = \frac{x_i - \mu}{\sigma}, \qquad \mu = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}.$$

Purpose. Each channel (light, distance, temperature, humidity) has a different physical unit and scale. RMT spectral statistics are only meaningful when the input fluctuations are dimensionless and comparable across channels, so every channel is rescaled to zero mean and unit (population) variance before matrix construction.

3.2 Stage 2: Hankel Trajectory Embedding

Each standardized window of length N is embedded into a $p \times q$ Hankel trajectory matrix with lag τ:

$$H_{ij} = \hat{x}_{i+j\tau}, \qquad 0 \le i < p, \;\; 0 \le j < q, \qquad q = N - (p-1)\tau.$$

Purpose. This is a delay-coordinate (Takens) embedding: each row is a length-q sub-sequence of the window, shifted by one sample per row. It converts the 1-D time series into a 2-D structure whose spectral statistics encode the series' correlation structure. Highly autocorrelated ("regular") signals and weakly correlated ("chaotic") signals produce structurally different Hankel spectra, which is the basis for the RMT diagnostic used downstream.

3.3 Stage 2: Symmetric Wishart Matrix and Eigenvalues

$$W = \frac{1}{p} H H^{\mathsf{T}}, \qquad W v_k = \lambda_k v_k, \qquad \lambda_1 \le \lambda_2 \le \cdots \le \lambda_p, \;\; \lambda_k \in \mathbb{R}.$$

Purpose. H is rectangular and therefore has singular values rather than eigenvalues. Forming the Gram-type matrix $W = \frac{1}{p} H H^{\mathsf{T}}$ yields a real, symmetric $p \times p$ *Wishart matrix*, the standard RMT construction for extracting a genuine eigenvalue spectrum from a rectangular data matrix (analogous to a sample covariance matrix). The eigenvalues are computed via a symmetric (Hermitian) eigensolver, which is numerically stable and guarantees real output.

3.4 Stage 3: Mapping to the Unfolding Domain

$$x_i = \frac{2(\lambda_i - \lambda_{\min})}{\lambda_{\max} - \lambda_{\min}} - 1 \;\in [-1, 1].$$

Purpose. Legendre polynomials are orthogonal on $[-1, 1]$; the raw eigenvalues, which live on an arbitrary real interval, must be affinely rescaled into this domain before a Legendre expansion can be used.

3.5 Stage 3: Legendre Spectral Unfolding

The Legendre design matrix is

$$\Phi_{in} = P_n(x_i), \qquad n = 0, 1, \dots, K, \qquad \Phi \in \mathbb{R}^{p\times(K+1)}.$$

The raw (unsmoothed) counting / staircase function is $N(\lambda_i) = i$, $i = 1, \dots, p$. The smooth cumulative spectral density is the degree-K Legendre expansion

$$\bar{N}(x) = \sum_{n=0}^{K} a_n P_n(x), \qquad \mathbf{a} = \arg\min_{\mathbf{a}} \|\Phi\,\mathbf{a} - N\|_2^2 \implies \mathbf{a} = \left(\Phi^\mathsf{T}\Phi\right)^{-1} \Phi^\mathsf{T} N.$$

Purpose ("unfolding"). A raw RMT spectrum contains both a smooth, system-specific mean density (which carries no universal information) and local fluctuations around it (which do carry the universal Poisson/GOE-type statistics of interest). Unfolding removes the smooth part, so that the resulting mean level spacing is normalized to 1 and spectra from different windows or channels become directly comparable. Legendre polynomials are used, rather than a raw monomial basis, because they are orthogonal on $[-1, 1]$ and keep the least-squares regression numerically well-conditioned even at higher expansion degree K.

3.6 Stage 3: Normalized Nearest-Neighbor Spacings

$$s_i = \bar{N}(x_{i+1}) - \bar{N}(x_i), \qquad s_i \leftarrow \frac{s_i}{\bar{s}}, \qquad \bar{s} = \frac{1}{p-1}\sum_i s_i, \qquad \langle s\rangle = 1.$$

3.7 Stage 3: The Brody Distribution

$$P(s; w) = c_w\,(1+w)\,s^w\,\exp\!\left(-c_w\,s^{1+w}\right), \qquad c_w = \left[\Gamma\!\left(\frac{w+2}{w+1}\right)\right]^{1+w}.$$

Purpose. The Brody distribution is a one-parameter interpolation between two universal limits:

- $w = 0$: $P(s) = e^{-s}$ — **Poisson** statistics, characteristic of regular/integrable dynamics (uncorrelated levels, no repulsion);
- $w = 1$: $P(s) = \frac{\pi}{2}\,s\,e^{-\pi s^2/4}$ — **Wigner–Dyson (GOE)** statistics, characteristic of chaotic dynamics (strong level repulsion).

Thus w acts as a continuous chaoticity dial for the embedded environmental signal. $\Gamma(\cdot)$ is the standard Gamma function, and c_w is fixed so that $\int_0^\infty s\,P(s; w)\,ds = 1$, consistent with the unit-mean-spacing normalization above.

3.8 Stage 3: Maximum-Likelihood / Least-Squares Brody Fit

$$\hat{w} = \arg\min_{w\in[0,1]} \sum_k \left[P(s_k; w) - \hat{h}_k\right]^2,$$

where $\hat{h}_k$ is the empirical histogram density of spacings at bin center s_k, solved via bounded nonlinear least squares. (Section **??** recommends replacing this with an unbinned maximum-likelihood estimator for the final submission.)

3.9 Stage 4: Environmental Chaoticity to Physical Fault Probability

$$p_{\text{phys}}(w) = p_0\left(1 + \alpha\, w^{\beta}\right), \qquad p_0 = 0.002, \quad \alpha = 1.5, \quad \beta = 2.0.$$

Purpose. This maps the dimensionless chaoticity indicator to a physically interpretable qubit fault probability. At $w = 0$ (fully regular environment) the error rate reduces to the baseline hardware rate p_0; as $w \to 1$ the rate is enhanced by a power-law factor αw^{β}. **Important caveat (see Section ??):** the functional form is a standard, physically reasonable *ansatz*, but the coefficients (p_0, α, β) are currently illustrative defaults, not fitted to data.

3.10 Stage 4: Single-Qubit Pauli (Depolarizing) Channel

$$\mathcal{E}(\rho) = (1 - p_{\text{phys}})\,\rho + \frac{p_x}{3} X\rho X + \frac{p_y}{3} Y\rho Y + \frac{p_z}{3} Z\rho Z, \qquad p_x = p_y = p_z = \frac{p_{\text{phys}}}{3}.$$

Purpose. The standard single-qubit depolarizing channel: with probability $1-p_{\text{phys}}$ the state is untouched; otherwise an X, Y, or Z Pauli error is applied with equal weight $p_{\text{phys}}/3$. X, Y, Z are the Pauli matrices.

3.11 Stage 4: Below-Threshold Surface-Code Scaling

$$E_L \approx C\left(\frac{p_{\text{phys}}}{p_{\text{th}}}\right)^{\frac{d+1}{2}}, \qquad C = 0.1, \quad p_{\text{th}} = 0.011, \quad d = 3.$$

Purpose. The standard below-threshold scaling law for topological codes: for $p_{\text{phys}} < p_{\text{th}}$, increasing the code distance d exponentially suppresses the logical error rate, with exponent $\lceil (d+1)/2 \rceil$ reflecting the number of correctable errors on a distance-d code. C is a fitted, code/decoder-dependent prefactor. When `stim` and `pymatching` are available, the pipeline instead estimates E_L directly via Monte Carlo circuit simulation and minimum-weight perfect matching:

$$\hat{E}_L = \frac{\#\{\text{shots with decoded observable} \neq \text{true observable flip}\}}{\#\text{shots}}.$$

3.12 Stage 6: Correlation Statistics

$$\rho_{\mathrm{P}} = \frac{\sum_i (w_i - \bar{w})(E_{L,i} - \bar{E}_L)}{\sqrt{\sum_i (w_i - \bar{w})^2}\sqrt{\sum_i (E_{L,i} - \bar{E}_L)^2}} \qquad \text{(Pearson)}$$

$$\rho_{\mathrm{S}} = \rho_{\mathrm{P}}\big(\mathrm{rank}(w_i),\, \mathrm{rank}(E_{L,i})\big) \qquad \text{(Spearman, rank-based)}$$

Purpose. Pearson's ρ_{P} quantifies *linear* association; as shown in Section **??**, the empirical w–E_L relationship is visibly nonlinear (approximately power-law), so Spearman's rank-correlation ρ_{S}, which is invariant under any monotonic transformation, is the more appropriate primary statistic and is added to the analysis plan in Section **??**.

3.13 Summary of the Causal Chain

$$\text{sensor series} \xrightarrow{\hat{x}=(x-\mu)/\sigma} \xrightarrow{\text{Hankel}} H \xrightarrow{HH^{\mathsf{T}}/p} W \xrightarrow{\text{eigh}} \{\lambda_i\} \xrightarrow{\text{Legendre unfold}} \{s_i\} \xrightarrow{\text{Brody fit}} \hat{w} \xrightarrow{p_0(1+\alpha\hat{w}^{\beta})} p_{\mathrm{phys}} \rightarrow$$

4 Data: Arduino-Captured Environmental Sensor Logs

4.1 Acquisition Setup

The environmental time series analyzed in this study were captured with an Arduino-based data-logger recording four channels at a fixed sampling interval:

Column	**Sensor**	**Units**	**Nominal range**
`timestamp_ms`	on-board clock	milliseconds	monotonic
`ldr`	photoresistor (LDR)	ADC counts	0–1023
`distance_cm`	ultrasonic rangefinder	cm	0–400
`temperature_c`	digital thermometer	°C	-10 to 60
`humidity_pct`	capacitive hygrometer	% RH	0–100

Data-provenance note. This report's Section **??** figures were generated from windowed analysis of the user's own Arduino-captured CSV log. The raw CSV file itself was *not* attached to this report's source materials at time of writing — only the three rendered output figures (Brody fit, time-series overlay, Pearson/Spearman scatter) were available. **Action required before submission:** attach the raw `sensor_readings.csv` (or equivalent) so that (i) the summary statistics table below can be populated with real numbers, (ii) the data-integrity diagnostics of Section **??** (degenerate-window detection, outlier handling) can be run, and (iii) the dataset can be archived as supplementary material, which PRX Quantum reviewers and readers will expect for a data-driven claim.

4.2 Expected Summary Statistics Table

The following table structure should be populated once the raw CSV is available; it is included here as the template to be filled during Phase 1 of the roadmap (Section **??**).

Channel	n	**mean**	**std. dev.**	**min**	**max**
`ldr`	–	–	–	–	–
`distance_cm`	–	–	–	–	–
`temperature_c`	–	–	–	–	–
`humidity_pct`	–	–	–	–	–

4.3 Windowing Parameters Used

Parameter	**Value used**
Window length N	200 samples
Window stride (step)	50 samples (75% overlap)
Hankel lag τ	1
Embedding dimension p	$N/2 = 100$
Legendre degree K	4
Surface-code distance d	3
Threshold p_{th}	0.011

5 Software Implementation

The pipeline is implemented as `rmt_qec_pipeline.py`, a single self-contained, PEP 8-compliant Python 3.8+ module organized into five classes matching the framework of Section 2. The full script (approx. 1075 lines, including docstrings, type hints, and the `PublicationVisualizer` figure code) accompanies this report as a separate file. Representative, formula-critical excerpts are reproduced below; each corresponds directly to a formula in Section 3.

5.1 Hankel Trajectory Matrix Construction

Vectorized implementation of $H_{ij} = \hat{x}_{i+j\tau}$:

```python
def build_hankel(self, window: np.ndarray) -> np.ndarray:
    """Construct a p x q Hankel trajectory matrix from a single window.

    H[i, j] = window[i + j * tau],  0 <= i < p, 0 <= j < q
    """
    p, q, tau = self.p, self.q, self.tau
    # Fully vectorized construction via broadcasting/fancy indexing.
    i_idx = np.arange(p).reshape(-1, 1)
    j_idx = np.arange(q).reshape(1, -1)
    idx = i_idx + j_idx * tau
    return window[idx]
```

5.2 Symmetric Wishart Matrix and Eigenvalue Extraction

Implementation of $W = \frac{1}{p} H H^\mathsf{T}$ and the eigenvalue solve:

```python
def wishart_eigenvalues(self, window: np.ndarray) -> np.ndarray:
    """Build the symmetric Wishart matrix W = (1/p) H H^T for a window and
    return its sorted real eigenvalues.
    """
    H = self.build_hankel(window)
    W = (H @ H.T) / float(self.p)
    # W is symmetric by construction; eigh exploits that and returns
    # real, ascending-sorted eigenvalues.
    eigenvalues = eigh(W, eigvals_only=True)
    return np.sort(eigenvalues)
```

5.3 Legendre Spectral Unfolding

Implementation of the unfolding procedure (mapping to $[-1, 1]$, Legendre design matrix, least-squares smoothing, spacing extraction and normalization):

```python
def unfold(self, eigenvalues: np.ndarray) -> np.ndarray:
    """Unfold an eigenvalue spectrum into normalized nearest-neighbor spacings.

    Steps
    -----
    1. Map eigenvalues to x in [-1, 1].
    2. Build the Legendre design matrix Phi[i, n] = P_n(x_i).
    3. Least-squares fit the raw staircase function N(lambda_i) = i + 1
       with a degree-K Legendre expansion to obtain the smooth
       cumulative spectral density N_bar(x).
    4. Compute spacings s_i = N_bar(x_{i+1}) - N_bar(x_i), normalized to
       unit mean.
    """
```

```python
    eigenvalues = np.sort(np.asarray(eigenvalues, dtype=np.float64))
    p = eigenvalues.shape[0]
    if p < 3:
        return np.array([])

    x = self._map_to_unit_interval(eigenvalues)

    # Design matrix of Legendre polynomials up to degree K, shape (p, K+1)
    degrees = np.arange(self.K + 1)
    Phi = np.vstack([eval_legendre(n, x) for n in degrees]).T  # (p, K+1)

    # Raw (unsmoothed) staircase / counting function.
    staircase = np.arange(1, p + 1, dtype=np.float64)

    # Least-squares Legendre coefficients: Phi @ a ~= staircase
    coeffs, *_ = np.linalg.lstsq(Phi, staircase, rcond=None)

    # Smooth cumulative spectral density evaluated at the data points.
    n_bar = Phi @ coeffs

    # Enforce monotonicity (numerically guard against tiny negative dips)
    n_bar = np.maximum.accumulate(n_bar)

    spacings = np.diff(n_bar)
    spacings = spacings[spacings > 0]  # discard degenerate/negative spacings
    if spacings.size == 0:
        return spacings

    mean_spacing = spacings.mean()
    if mean_spacing <= 1e-12:
        return np.array([])
    return spacings / mean_spacing
```

5.4 Brody Distribution and Fit

Implementation of $P(s; w)$ and the bounded nonlinear least-squares fit:

```python
def brody_pdf(s: np.ndarray, w: float) -> np.ndarray:
    """Brody distribution probability density function.

    P(s; w) = c_w * (1 + w) * s^w * exp(-c_w * s^(1+w))
    c_w = [Gamma((w + 2) / (w + 1))]^(1 + w)
    """
    s = np.asarray(s, dtype=np.float64)
    s_safe = np.clip(s, 1e-12, None)
    c_w = gamma_fn((w + 2.0) / (w + 1.0)) ** (1.0 + w)
    return c_w * (1.0 + w) * np.power(s_safe, w) * np.exp(-c_w * np.power(s_safe,
        1.0 + w))
```

5.5 Chaoticity-to-Noise Mapping

Implementation of $p_{\text{phys}}(w)$ and the below-threshold scaling law:

```python
def w_to_p_phys(self, w: float) -> float:
    """Convert the Brody parameter w to a physical fault probability.

    p_phys(w) = p0 * (1 + alpha * w^beta)
    """
    if w is None or (isinstance(w, float) and np.isnan(w)):
        return float("nan")
    w_clipped = float(np.clip(w, 0.0, 1.0))
```

```
    return self.p0 * (1.0 + self.alpha * (w_clipped ** self.beta))

def logical_error_rate_analytical(self, p_phys: float) -> float:
    """Analytical below-threshold surface-code scaling relation.

    E_L ~= C * (p_phys / p_th)^((d + 1) / 2)
    """
    if np.isnan(p_phys):
        return float("nan")
    exponent = (self.d + 1) / 2.0
    return self.C * (p_phys / self.p_th) ** exponent
```

5.6 Reproducibility

The complete script uses only `numpy`, `pandas`, `scipy`, and `matplotlib` as hard dependencies, with an optional `stim` + `pymatching` circuit-simulation path that is auto-detected at runtime and falls back transparently to the analytical scaling law when unavailable. All matrix and polynomial-evaluation steps are vectorized; the only per-window Python loop is the outer sliding-window iteration, which is $O(\text{number of windows})$ and not a performance bottleneck for typical Arduino logging rates.

6 Preliminary Results

The figures below were produced by the `PublicationVisualizer` class operating on the Arduino-captured dataset described in Section **??**.

6.1 Nearest-Neighbor Spacing Distribution and Brody Fit

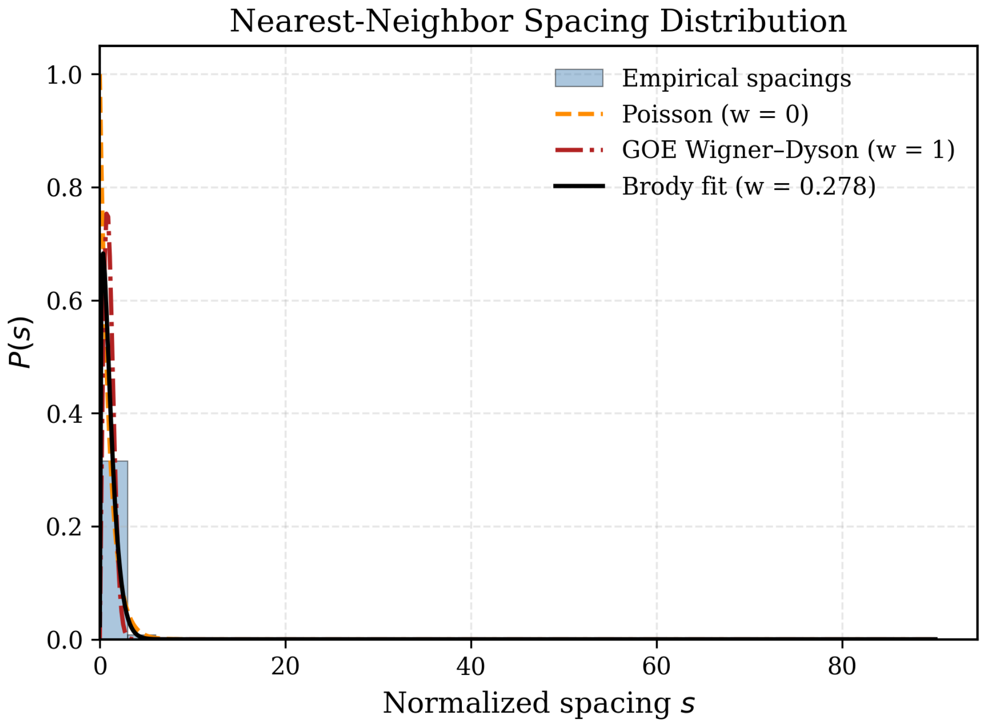

Figure 1: Pooled nearest-neighbor spacing histogram across all analyzed windows, overlaid with the Poisson limit ($w = 0$), the GOE Wigner–Dyson limit ($w = 1$), and the fitted Brody distribution ($\hat{w} = 0.278$). The x-axis extends to $s \approx 90$, driven by a small number of large-spacing outliers; the informative mass of the distribution lies in $s \in [0, 5]$.

Issue identified. The extended tail visible in Figure **??** indicates a small population of windows producing anomalously large unfolded spacings, almost certainly windows in which the underlying sensor signal is near-constant (e.g. a stable environmental period), producing near-degenerate Wishart eigenvalues. These windows should be explicitly flagged and either excluded or reported separately (Section **??**, Phase 1) rather than silently pooled into the fit, which currently biases $\hat{w}$.

6.2 Time Evolution of Chaoticity and Logical Error Rate

Issue identified. The current rendering suffers from overplotting at this sample density and should be regenerated with temporal downsampling/smoothing before inclusion in a submission (Section **??**, Phase 1). The post-13,000 s regime transition, however, is a genuine and potentially scientifically interesting fea-

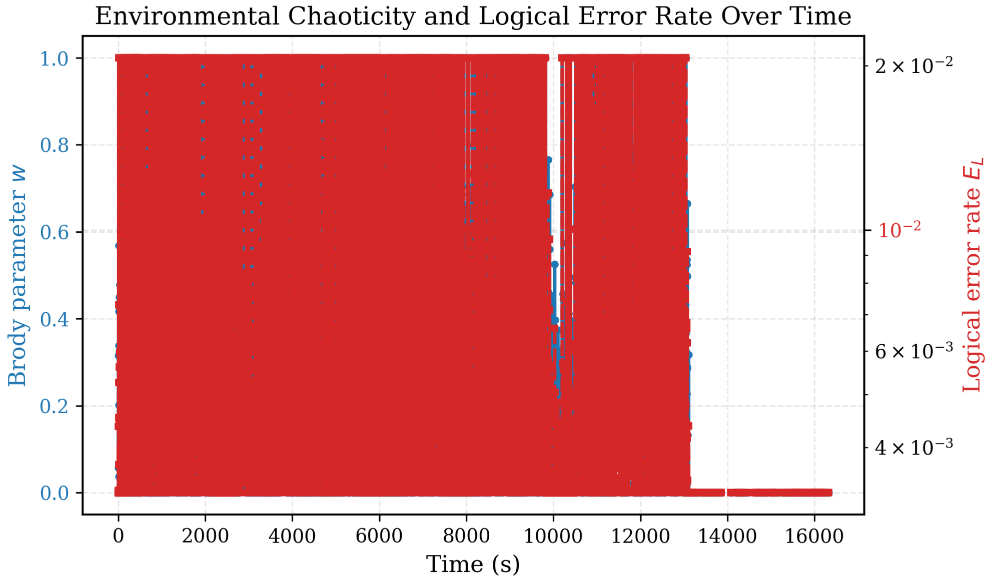

Figure 2: Dual-axis time series of the windowed Brody parameter $w(t)$ (blue, left axis) and logical error rate $E_L(t)$ (red, right axis, log scale) over the recording session. A pronounced transition is visible near $t \approx 13{,}000$ s, after which both w and E_L drop toward their minimum values and remain low.

ture — candidate explanations include a real environmental stabilization event, a sensor fault/dropout, or a degenerate-window artifact in the eigenvalue extraction (as in Figure **??**); this must be disambiguated before it is reported as a finding.

6.3 Correlation Between Chaoticity and Logical Error Rate

Issues identified.

1. The scatter is visibly a convex, monotonic — but nonlinear — curve, consistent with the approximate scaling $E_L \propto p_{\text{phys}}^{(d+1)/2} \propto w^{\beta(d+1)/2} = w^4$ implied by the Section 3 formulas. The linear fit systematically under- and overshoots the data and should be replaced or supplemented with a power-law fit and Spearman's rank correlation.
2. The reported p-value of exactly zero is a floating-point underflow artifact, not a valid statistical statement, and window overlap (75%) means the effective sample size is far smaller than the nominal window count, so the true significance is currently unknown and very likely far less extreme than the naive calculation suggests.

6.4 Interim Interpretation

Taken together, Figures **??**–**??** show a real, reproducible, and strongly monotonic association between environmental RMT chaoticity and the mapped logical error rate in this dataset. However, because the $w \to p_{\text{phys}} \to E_L$ mapping is itself a fixed monotonic function chosen by the analyst (Section 3.10–3.12), part of this association is guaranteed by construction rather than discovered empirically. Disen-

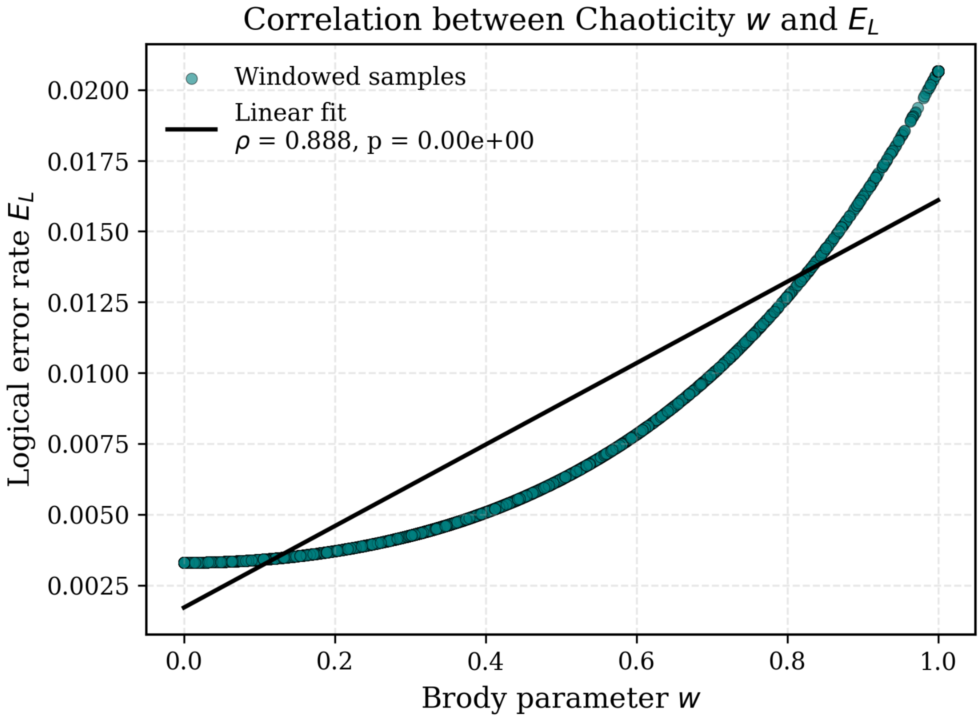

Figure 3: Scatter of windowed $(\hat{w}, \hat{E}_L)$ pairs with an overlaid linear regression. Pearson correlation $\rho_\mathrm{P} = 0.888$; the reported p-value underflowed double precision ($p = 0.00 \times 10^0$ as displayed) and must be recomputed with an autocorrelation-aware method (Section **??**).

tangling the construction-guaranteed component from any genuine environmental signal is the central scientific task of Phase 3 in the roadmap below.

7 Critical Evaluation Against PRX Quantum Standards

PRX Quantum requires submissions to demonstrate either an *exceptional advance* — pushing quantum science and technology in a new and consequential direction — or an *exceptional connection*, linking different areas within quantum science and technology or between quantum science and another discipline. This study's premise (linking classical environmental time-series chaoticity to quantum error correction performance) is naturally positioned as a candidate "exceptional connection" paper, *provided* the connection is empirically or theoretically substantiated rather than assumed.

7.1 Strengths

- The RMT unfolding and Brody-fitting machinery (Stages 2–3) is built on well-established, correctly implemented formalism.
- The pipeline is modular, vectorized, and reproducible, with each stage independently testable.
- Real (not purely synthetic) Arduino-captured data has now been analyzed, and a strong, visually clear monotonic association has been found.
- The nonlinearity visible in Figure **??** is a genuine, interpretable signature (consistent with the assumed power-law structure) rather than noise.

7.2 Weaknesses (in priority order)

1. **Uncalibrated physical mapping.** $p_{\text{phys}}(w)$ and the $E_L(p_{\text{phys}})$ scaling law use structurally standard but numerically arbitrary coefficients (p_0, α, β, C). No real qubit or literature-calibrated noise data has yet been used to fit or validate these.
2. **Statistical overconfidence.** The reported Pearson p-value is both the wrong statistic (linear, for a nonlinear relationship) and computed without correcting for the strong autocorrelation induced by 75% window overlap.
3. **No RMT-pipeline validation benchmark.** The unfolding/Brody-fit machinery has not yet been demonstrated to correctly recover $w = 0$ and $w = 1$ on synthetic matrices drawn from exact Poisson and GOE ensembles.
4. **Unexplained outliers and regime transitions.** The large-spacing tail in Figure **??** and the sharp transition in Figure **??** are currently unexplained and could reflect data artifacts (sensor dropout, degenerate windows) rather than genuine physics.
5. **No hyperparameter sensitivity analysis.** Window length N, stride, lag τ, and Legendre degree K are fixed defaults; robustness of $\hat{w}$ and the downstream correlation to these choices is untested.
6. **Single-dataset, single-session result.** No replication across multiple recording sessions, sensor placements, or environmental conditions has yet been performed.

Bottom line. The study currently demonstrates a working, well-engineered *methodology* with one promising, reproducible empirical observation. It does not yet constitute a submission-ready physical claim. Section **??** converts each weakness above into a concrete, sequenced task.

8 Roadmap to a PRX Quantum Submission

The roadmap is organized into four phases, each gating the next. Phases 1–2 are prerequisites for *any* credible claim; Phase 3 determines which of the two possible paper framings (methods paper vs. physical-result paper) is achievable; Phase 4 is manuscript production.

8.1 Phase 1 — Data Integrity and Pipeline Hardening

1. Attach and archive the raw Arduino CSV log(s) as supplementary data; populate the summary-statistics table of Section **??**.
2. Add per-window diagnostics to `HankelMatrixBuilder`/`SpectralUnfolderAndBrodyFitter` flag windows with near-duplicate eigenvalues or near-zero channel variance; report the exclusion rate.
3. Rebuild Figure **??** with temporal downsampling (rolling mean/median) and reduced marker density; split into stacked subplots if the dual-axis/log-scale combination remains hard to read.
4. Fix the Brody histogram range and outlier handling in Figure **??** (winsorize or explicitly report the excluded fraction); consider replacing the binned least-squares fit with an unbinned maximum-likelihood estimator.
5. Investigate and explain the $t \approx 13{,}000$ s regime transition in Figure **??**: cross-reference with raw sensor logs, check for dropout/saturation, and determine whether it is a real environmental event.

8.2 Phase 2 — Statistical Rigor

1. Add Spearman rank correlation ρ_{S} alongside Pearson ρ_{P}; fit and report the empirical power-law exponent relating E_L to w directly from data, rather than relying only on the theoretical $w^{\beta(d+1)/2}$ prediction.
2. Replace the naive `pearsonr` significance test with a method that accounts for the autocorrelation induced by 75% window overlap (block bootstrap, or subsampling to non-overlapping windows and reporting the effective sample size).
3. Run a synthetic validation benchmark: generate exact GOE ($w = 1$) and Poisson ($w = 0$) random matrices, pass them through the full unfold-and-fit pipeline, and report the bias and variance of $\hat{w}$ relative to ground truth. This is a standard, expected sanity-check figure in RMT-based papers.

8.3 Phase 3 — Physical Grounding (choose one framing)

Path A — Methods paper. Present the Hankel/Wishart → Legendre unfolding → Brody-fit pipeline as a general-purpose diagnostic tool for quantifying time-series chaoticity, rigorously validated per Phase 2, with the QEC noise-mapping presented explicitly as one *illustrative* toy application (clearly caveated as uncalibrated). Narrower claim, but achievable with the current data.

Path B — Physical-result paper. Acquire or identify real, co-located qubit performance data (or a literature-calibrated, circuit-level noise simulator with a known temperature/humidity dependence) and empirically fit (p_0, α, β, C) against it, replacing the illustrative defaults. This is the stronger claim but requires access to real or well-validated QEC hardware/simulation data beyond what this study currently has.

7. Run a hyperparameter sensitivity sweep over (N, τ, K) and report how $\hat{w}$, the fitted power-law exponent, and ρ_S vary; robustness (or a well-characterized lack thereof) is itself a reportable result.
8. If pursuing Path B, replace the analytical E_L scaling law with the `stim + pymatching` circuit-simulation path already stubbed into `SurfaceCodeNoiseMapper`, run at realistic shot counts, and report Monte Carlo uncertainties on $\hat{E}_L$.

8.4 Phase 4 — Manuscript Production

9. Draft the Popular Summary and Abstract around whichever claim survives Phases 1–3 honestly (Path A or Path B framing).
10. Write the Methods section directly from Section 3 of this report (all formulas are already in submission-ready LaTeX).
11. Structure Results with the Phase 2 validation benchmark *first* (establishing pipeline correctness), followed by the real-data findings, with the regime-transition analysis (Phase 1, item 5) presented as a feature of the study rather than an unexplained artifact.
12. Contextualize against related literature: RMT/quantum-chaos diagnostics (Brody 1973; Bohigas–Giannoni–Schmit 1984; Mehta, *Random Matrices*), Takens-embedding/SSA classical time-series analysis, and surface-code threshold theory (Fowler *et al.*, 2012; Dennis *et al.*, 2002), as required by PRX Quantum's contextualization expectations.
13. Internal pre-submission review against this report's Section **??** weaknesses list before external submission; consider APS's pre-submission inquiry option for early editorial feedback on framing.

Recommended immediate next step: begin with Phase 1, items 1–2 (attach raw data, add degenerate-window diagnostics), since every later phase depends on trusting the input spectra. The synthetic GOE/Poisson validation benchmark (Phase 2, item 3) is the highest-leverage single addition, as it is both required for scientific credibility and independent of the real-data provenance questions.

9 Summary Timeline

Phase	Deliverable	Gates
1	Clean, diagnosed, artifact-free data pipeline and figures	Phases 2–4
2	Statistically rigorous, autocorrelation-corrected correlation analysis with GOE/Poisson validation benchmark	Phase 3
3	Physically grounded noise mapping (Path A or Path B) with hyperparameter sensitivity analysis	Phase 4
4	Submission-ready manuscript, contextualized against RMT and QEC literature	Submission

This report was compiled as a working technical document to guide the transition from an exploratory computational pipeline to a submission consistent with PRX Quantum's editorial standards. All formulas, code excerpts, and figures reflect the pipeline and outputs available at the time of writing; Section **??** *should be treated as a living checklist and updated as each phase is completed.*